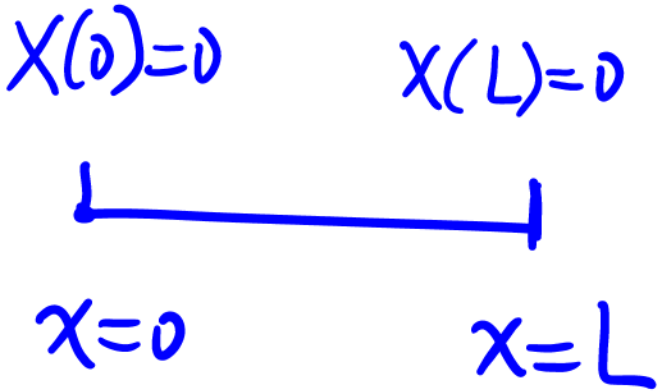
Solve $X''(x) + \lambda X(x) = 0$



$$X(0) = X(L) = 0$$

$$\lambda \in \mathbb{R} \qquad L > 0$$

characteristic equation

$$\Lambda^2 + \lambda = 0$$

① $\lambda < 0$ $\quad \Lambda^2 = -\lambda \quad \Lambda_{1,2} = \pm\sqrt{-\lambda}$

$$X(x) = c_1 e^{\sqrt{-\lambda}\, x} + c_2 e^{-\sqrt{-\lambda}\, x}$$

$$X(0) = 0 \implies c_1 + c_2 = 0$$

$$X(L) = 0 \implies c_1 e^{\sqrt{-\lambda}\, L} + c_2 e^{-\sqrt{-\lambda}\, L} = 0$$

We have $c_1 = c_2 = 0$. Hence, $\underline{X(x) = 0}$

② $\lambda = 0$ $\quad \Lambda^2 = 0 \implies \Lambda_{1,2} = 0$

$$X(x) = c_1 e^{\Lambda x} + c_2 x e^{\Lambda x}$$

$$= c_1 + c_2 x \xrightarrow{X(0) = X(L) = 0} c_1 = c_2 = 0$$

③ $\lambda > 0 \quad \Lambda^2 + \lambda = 0 \quad \Lambda_{1,2} = \pm\sqrt{\lambda}\, i$

$$X(x) = C_1 \cos\sqrt{\lambda}\, x + C_2 \sin\sqrt{\lambda}\, x$$

From $X(0) = 0$, we have $C_1 = 0$

$$\left.\begin{array}{l} X(L) = 0 \\ C_1 = 0 \end{array}\right\} \Rightarrow C_2 \sin\sqrt{\lambda}\, L = 0$$

$$C_2 \sin\sqrt{\lambda}\, L = 0 \Rightarrow \begin{cases} C_2 = 0 & \text{Trivial Solution} \\ \sin\sqrt{\lambda}\, L = 0 \end{cases}$$

$$\sin \sqrt{\lambda}\, L = 0$$

$$\Rightarrow \sqrt{\lambda}\, L = k\pi \qquad k = 1, 2, \cdots$$

$$\Rightarrow \sqrt{\lambda} = \frac{k\pi}{L}$$

$$X'(x) = C_2 \frac{k\pi}{L} \cos \frac{k\pi}{L} x$$

$$\Rightarrow \lambda = \frac{k^2 \pi^2}{L^2}$$

$$X''(x) = -C_2 \frac{k^2 \pi^2}{L^2} \sin \frac{k\pi}{L} x$$

$$X(x) = C_2 \sin \frac{k\pi}{L} x$$

$$\lambda_k = \frac{k^2 \pi^2}{L^2} \qquad k = 1, 2, \cdots$$

$$X_k(x) = \sin \frac{k\pi}{L} x$$

Then $X(x) = \sum_{k=1}^{\infty} \sin \frac{k\pi}{L} x$

is also a solution only if the RHS is convergent

MH4110 Partial Differential Equations

Chapter 4 - Boundary problems

Outline

- Initial Value Problems and Initial Boundary Value Problems
- Boundary Value Problems
- Separation of Variables: Homogeneous Dirichlet conditions
  - *Wave equations*
  - *Heat equations*
- Separation of Variables: Homogeneous Neumann conditions
  - *Wave equations*
  - *Heat equations*

Initial Value/Cauchy Problems

Wave equation on the whole line:

$$u_{tt} = c^2 u_{xx}, \quad x \in \mathbb{R} = (-\infty, \infty), \quad t > 0,$$
$$u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad x \in \mathbb{R},$$

where ϕ, ψ are any given functions.

Initial Value/Cauchy Problems

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$$u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad x \in \mathbb{R},$$

where ϕ, ψ are any given functions.

Solution formula–d'Alembert formula:

$$u(x,t) = \frac{1}{2}\big[\phi(x+ct) + \phi(x-ct)\big] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)ds.$$

Imagine the picture of the solution with $\psi = 0$.

Initial Value/Cauchy Problems (Cont'd)

Heat equation on the whole line

$$
\begin{aligned}
&u_t = ku_{xx}, \quad x \in \mathbb{R} = (-\infty, \infty), \quad t > 0, \\
&u(x, 0) = \phi(x), \quad x \in \mathbb{R},
\end{aligned}
$$

where ϕ is a given function.

Initial Value/Cauchy Problems (Cont'd)

Heat equation on the whole line

$$u_t = ku_{xx}, \quad x \in \mathbb{R} = (-\infty, \infty), \quad t > 0,$$
$$u(x, 0) = \phi(x), \quad x \in \mathbb{R},$$

where ϕ is a given function.

Solution formula:

$$u(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \exp\Big(- \frac{(x-y)^2}{4kt} \Big) \phi(y) dy.$$

Let $S(x,t) = \frac{1}{\sqrt{4\pi kt}} e^{-\frac{x^2}{4kt}}$. Then we can write the solution as a convolution:

$$u(x,t) = S(\cdot, t) * \phi.$$

Initial Boundary Value Problems

Wave equation on the half line with Dirichlet boundary conditions

$$
\begin{aligned}
&v_{tt} - c^2 v_{xx} = 0, \quad 0 < x < \infty, \quad t > 0, \\
&v(x,0) = \phi(x), \quad v_t(x,0) = \psi(x), \quad \text{(initial condition at } t = 0), \\
&v(0,t) = 0 \quad \text{(boundary condition at } x = 0),
\end{aligned}
$$

where we assume that $\phi(0) = 0$ (consistent condition).

Initial Boundary Value Problems

Wave equation on the half line with Dirichlet boundary conditions

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\end{aligned}
$$

where we assume that $\phi(0) = 0$ (consistent condition).

Solution formula:

$$
v(x,t) = \begin{cases} \dfrac{1}{2}\left[\phi(x+ct) + \phi(x-ct)\right] + \dfrac{1}{2c}\displaystyle\int_{x-ct}^{x+ct} \psi(s)ds, & x > ct, \\ \dfrac{1}{2}\left[\phi(x+ct) - \phi(ct-x)\right] + \dfrac{1}{2c}\displaystyle\int_{ct-x}^{ct+x} \psi(s)ds, & 0 < x < ct. \end{cases}
$$

Initial Boundary Value Problems (Cont'd)

Wave equation on the half line with Neumann boundary conditions

$$w_{tt} - c^2 w_{xx} = 0, \quad 0 < x < \infty, \quad t > 0,$$
$$w(x,0) = \phi(x), \quad w_t(x,0) = \psi(x), \quad \text{(initial condition at } t = 0),$$
$$w_x(0,t) = 0 \quad \text{(boundary condition at } x = 0).$$

Initial Boundary Value Problems (Cont'd)

Wave equation on the half line with Neumann boundary conditions

$$w_{tt} - c^2 w_{xx} = 0, \quad 0 < x < \infty, \quad t > 0,$$
$$w(x,0) = \phi(x), \quad w_t(x,0) = \psi(x), \quad \text{(initial condition at } t = 0\text{)},$$
$$w_x(0,t) = 0 \quad \text{(boundary condition at } x = 0\text{)}.$$

Solution formula:

$$w(x,t) = \begin{cases} \dfrac{\phi(x+ct) + \phi(x-ct)}{2} + \dfrac{1}{2c}\displaystyle\int_{x-ct}^{x+ct} \psi(s)ds, & x > ct, \\[2ex] \dfrac{\phi(x+ct) + \phi(ct-x)}{2} + \dfrac{1}{2c}\left(\displaystyle\int_0^{ct-x} + \int_0^{x+ct}\right)\psi(s)ds, & 0 < x < ct. \end{cases}$$

Initial Boundary Value Problems (Cont'd)

Heat equation on the half line with Dirichlet boundary conditions

$$
\begin{aligned}
&v_t - kv_{xx} = 0, \quad 0 < x < \infty, \quad t > 0, \\
&v(x,0) = \phi(x), \quad \text{(initial condition at } t = 0), \\
&v(0,t) = 0 \quad \text{(boundary condition at } x = 0).
\end{aligned}
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Initial Boundary Value Problems (Cont'd)

Heat equation on the half line with Dirichlet boundary conditions

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\begin{aligned}
&v_t - kv_{xx} = 0, \quad 0 < x < \infty, \quad t > 0, \\
&v(x,0) = \phi(x), \quad \text{(initial condition at } t = 0), \\
&v(0,t) = 0 \quad \text{(boundary condition at } x = 0).
\end{aligned}
$$

Solution formula:

$$
v(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_0^\infty \Big[\exp\Big(-\frac{(x-y)^2}{4kt}\Big) - \exp\Big(-\frac{(x+y)^2}{4kt}\Big)\Big]\phi(y)dy.
$$

Initial Boundary Value Problems (Cont'd)

Heat equation on the half line with Neumann boundary conditions

$$
\begin{aligned}
&w_t - kw_{xx} = 0, \quad 0 < x < \infty, \quad t > 0, \\
&w(x,0) = \phi(x), \quad \text{(initial condition at } t = 0), \\
&w_x(0,t) = 0 \quad \text{(boundary condition at } x = 0).
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Initial Boundary Value Problems (Cont'd)

Heat equation on the half line with Neumann boundary conditions

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\end{aligned}
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Solution formula:

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w(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_0^\infty \Big[\exp\Big(-\frac{(x-y)^2}{4kt}\Big) + \exp\Big(-\frac{(x+y)^2}{4kt}\Big)\Big]\phi(y)dy.
$$

§4.1 Boundary value problems

The absence of boundaries is a big simplification, but most of physical problems have fixed boundaries. Therefore, it is necessary to consider boundary value problems.

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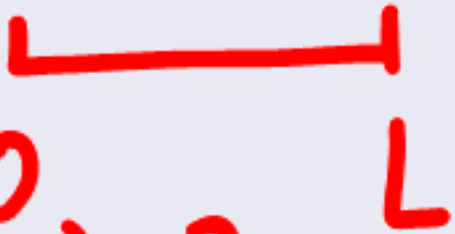
Wave equation with homogeneous Dirichlet boundary conditions

The wave equation

$$u_{tt} = c^2 u_{xx}, \quad 0 < x < l, \quad t > 0, \tag{1}$$

is subjected to the homogeneous boundary conditions

fixed boundary



$$u(0,t) = u(l,t) = 0, \quad t > 0, \tag{2}$$

$u(0,t)=0$

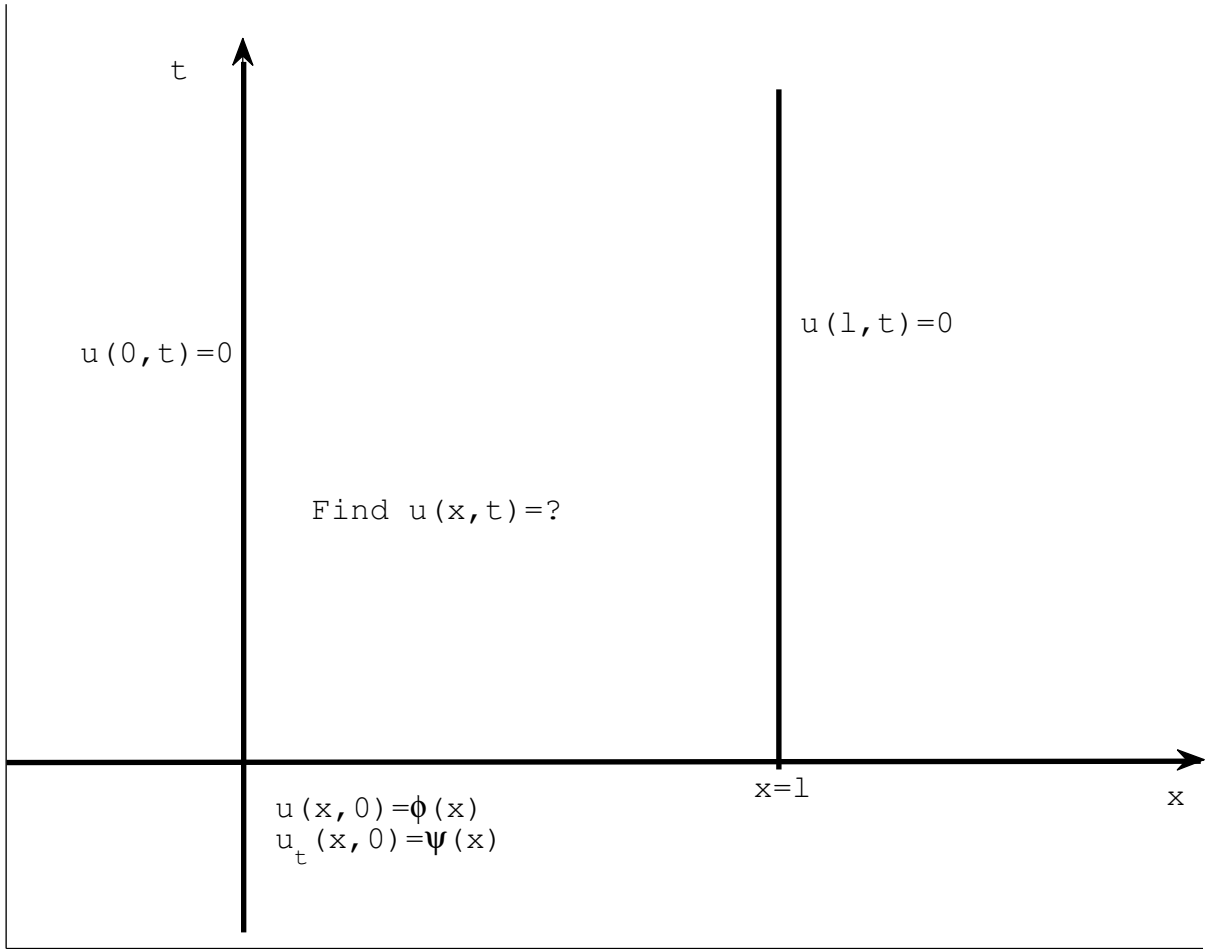
and the initial condition

$$u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad 0 < x < l, \tag{3}$$

where $c, l > 0$ are given constants, and ϕ, ψ are given functions.

§4.1 Boundary value problems (Cont'd)

Wave equations with Dirichlet boundary conditions on a finite interval



$u(0,t) = u(L,t) = 0$

Separation of Variables

Look for a **separated solution** of (39) of the form

$u(0,t) = X(0)T(t) = X(L)T(t) = u(L,t)$

$$u(x,t) = X(x)T(t), \tag{4}$$

$\Longrightarrow X(0) = X(L) = 0$

which should satisfy the initial and boundary conditions (2)-(3). This form will be substituted into (39) and result in ODEs for X and T, respectively. The method of splitting x and t dependency as in (4) is called **separation of variables**.

$u_{tt} = X(x)T''(t)$

$u_{xx} = X''(x)T(t)$

§4.1 Separation of Variables: The Dirichlet condition

Separation of Variables

Look for a **separated solution** of (39) of the form

$$u(x,t) = X(x)T(t), \tag{4}$$

which should satisfy the initial and boundary conditions (2)-(3). This form will be substituted into (39) and result in ODEs for X and T, respectively. The method of splitting x and t dependency as in (4) is called **separation of variables**.

We shall see that the applicability of this method essentially depends on the type of equations and solution domains (and type of boundary conditions sometimes).

Step I: Plugging (4) into (39) leads to

$$\underbrace{X(x)T''(t)}_{u_{tt}} = c^2 \underbrace{X''(x)T(t)}_{u_{xx}}. \tag{5}$$

$$-\frac{X(x)T''(t)}{c^2X(x)T(t)} = -\frac{X''(x)T(t)}{X(x)T(t)}$$

§4.1 Separation of Variables–Method

Step I: Plugging (4) into (39) leads to

$$X(x)T''(t) = c^2 X''(x)T(t). \tag{5}$$

Dividing both sides of the identity by $-c^2 X(x)T(t)$ to separate the variables, we get

$$-\frac{X''(x)}{X(x)} = -\frac{T''(t)}{c^2 T(t)} := \lambda. \tag{6}$$

§4.1 Separation of Variables–Method

Step I: Plugging (4) into (39) leads to $T''(t) + c^2\lambda T(t) = 0$

$$X(x)T''(t) = c^2 X''(x)T(t). \tag{5}$$

Dividing both sides of the identity by $-c^2 X(x)T(t)$ to separate the variables, we get

$$-\frac{X''(x)}{X(x)} = -\frac{T''(t)}{c^2 T(t)} := \lambda. \tag{6}$$

Two observations are in order. $\hookrightarrow X''(x) + \lambda X(x) = 0$

(i) λ must be a **constant**. As x and t are two independent variables, we find that

$X(0) = X(L) = 0$

$$\frac{\partial \lambda}{\partial t} = \frac{\partial}{\partial t}\left(-\frac{X''(x)}{X(x)}\right) = 0; \quad \frac{\partial \lambda}{\partial x} = \frac{\partial}{\partial x}\left(-\frac{T''(t)}{c^2 T(t)}\right) = 0.$$

Thus, λ is a constant.

§4.1 Separation of Variables–Method

Step I: Plugging (4) into (39) leads to

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Thus, λ is a constant.

(ii) We will shortly see that the boundary conditions force $\lambda > 0$ to be positive, so we put minus signs in (6).

§4.1 Separation of Variables–Method (Cont'd)

So let $\lambda = \beta^2$ with $\beta > 0$. Then we arrive at a pair of **separate** ODEs:

$$X''(x) + \beta^2 X(x) = 0; \quad T''(t) + c^2\beta^2 T(t) = 0. \tag{7}$$

§4.1 Separation of Variables–Method (Cont'd)

So let $\lambda = \beta^2$ with $\beta > 0$. Then we arrive at a pair of **separate** ODEs:

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Step II: Solve the ODEs (7) and find the value of λ (i.e., β) so that the solution satisfies the boundary conditions (2).

We started with the first ODE, and find that it has the **general solution**:

$$X(x) = C\cos\beta x + D\sin\beta x, \tag{8}$$

where C and D are two arbitrary constants.

§4.1 Separation of Variables–Method (Cont'd)

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To meet the boundary conditions (2), we require that

$$u(0,t) = X(0)T(t) = 0 = u(l,t) = X(l)T(t) \Rightarrow X(0) = X(l) = 0. \tag{9}$$

§4.1 Separation of Variables–Method (Cont'd)

So let $\lambda = \beta^2$ with $\beta > 0$. Then we arrive at a pair of **separate** ODEs:

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Note that $T(t) \equiv 0$ would result in the trivial solution $u(x,t) = 0$.

§4.1 Separation of Variables–Method (Cont'd)

Then, we find from (8) that

$$X(0) = C = 0, \quad X(l) = D \sin \beta l = 0. \tag{10}$$

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The solution with $D = 0$ will again lead to the trivial zero solution, so we consider the case when $\sin \beta l = 0$, implying that

$$\sin \beta l = 0 \quad \Rightarrow \quad \beta l = n\pi \quad \Rightarrow \quad \beta = \frac{n\pi}{l}, \tag{11}$$

where n could be any integer at this point.

§4.1 Separation of Variables–Method (Cont'd)

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where n could be any integer at this point.

Step III: Formulate the solution.
We find from the above that

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin \frac{n\pi x}{l}. \tag{12}$$

For each n, the pair (λ_n, X_n) is a solution of the **eigen-problem**:

$$X'' = -\lambda X, \quad X(0) = X(l) = 0. \tag{13}$$

§4.1 Separation of Variables–Method (Cont'd)

Now, we turn to the second equation of (7) with β given by (11). We find that the general solution is

$$T_n(t) = A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l}, \tag{14}$$

for each $n = 1, 2, 3, \cdots$, where A_n and B_n are arbitrary constants.

$$\left.\begin{aligned} &T''(t) + c^2 \lambda T(t) = 0 \\ &\lambda_n = \frac{n^2 \pi^2}{L^2} \end{aligned}\right\}$$

The roots of the charact. equation are

$$\pm \frac{n\pi c}{L} i$$

§4.1 Separation of Variables–Method (Cont'd)

Now, we turn to the second equation of (7) with β given by (11). We find that the general solution is

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for each $n = 1, 2, 3, \cdots$, where A_n and B_n are arbitrary constants.

Therefore, the boundary value problem (39)-(2) (we have not imposed the initial condition (3) yet) have infinite number of separated solutions:

$$u_n(x,t) = X_n(x)T_n(t) = \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l}\right) \sin \frac{n\pi x}{l}, \tag{15}$$

for all $n = 1, 2, 3, \cdots$.

§4.1 Separation of Variables–Method (Cont'd)

Based on the **principle of superposition**, the series

$$u(x,t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l} \right) \sin \frac{n\pi x}{l} \tag{16}$$

is also a solution of (39)-(2).

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 \\ u(0,t) = u(L,t) = 0 \end{cases}$$

§4.1 Separation of Variables–Method (Cont'd)

Based on the **principle of superposition**, the series

$$u(x,t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l} \right) \sin \frac{n\pi x}{l} \tag{16}$$

is also a solution of (39)-(2).

$$u_t = \sum_{n=1}^{\infty} \frac{n\pi c}{L} \left(-A_n \sin \frac{n\pi ct}{L} + B_n \cos \frac{n\pi ct}{L} \right) \sin \frac{n\pi x}{L}$$

Step IV: Use the initial condition (3) to determine A_n and B_n

$$\phi(x) = u(x,0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l},$$

$$\psi(x) = u_t(x,0) = \sum_{n=1}^{\infty} \frac{n\pi c}{l} B_n \sin \frac{n\pi x}{l}.$$

In Chapter 5, we shall use the Fourier series of ϕ and ψ to find A_n and B_n.

§4.1 Separation of Variables–Method (Cont'd)

Based on the **principle of superposition**, the series

$$u(x,t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l} \right) \sin \frac{n\pi x}{l} \tag{16}$$

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In Chapter 5, we shall use the Fourier series of ϕ and ψ to find A_n and B_n.

Question: Why $\lambda > 0$?

§4.1 Separation of Variables–Method (Cont'd)

Now, we come to this issue by showing that $\lambda \leq 0$ does not provide any meaningful solutions.

§4.1 Separation of Variables–Method (Cont'd)

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- **Case I: $\lambda = 0$.** In this case, the first equation of (6) becomes $X'' = 0$, so $X(x) = ax + b$. To meet the boundary conditions (9), we find $a = b = 0$ so $X(x) = 0$. The solution of the form (4) is trivial, so we reject $\lambda = 0$.

§4.1 Separation of Variables–Method (Cont'd)

Now, we come to this issue by showing that $\lambda \leq 0$ does not provide any meaningful solutions.

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- **Case II: $\lambda < 0$.** In this case, the first equation of (6) becomes

$$X''(x) + \lambda X(x) = 0 \quad \Rightarrow \quad X(x) = c_1 e^{-\sqrt{-\lambda}x} + c_2 e^{\sqrt{-\lambda}x}.$$

  By the boundary conditions $X(0) = X(l) = 0$, we find that

$$c_1 + c_2 = 0, \quad c_1 e^{-\sqrt{-\lambda}l} + c_2 e^{\sqrt{-\lambda}l} = 0.$$

  This 2-by-2 linear system with unknowns c_1, c_2 has only zero solution, i.e., $c_1 = c_2 = 0$. Thus, once again, we obtain trivial solution $u(x,t) = 0$, so we reject $\lambda < 0$ either.

§4.1 Separation of Variables–Method (Cont'd)

Theorem 1

The wave equation with homogeneous Dirichlet boundary conditions (39)-(3) has the following series solution:

$$u(x,t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l} \right) \sin \frac{n\pi x}{l}, \tag{17}$$

where A_n and B_n are to be determined by the initial condition (3).

§4.1 Separation of Variables–Method (Cont'd)

Theorem 1

The wave equation with homogeneous Dirichlet boundary conditions (39)-(3) has the following series solution:

$$u(x,t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l} \right) \sin \frac{n\pi x}{l}, \tag{17}$$

where A_n and B_n are to be determined by the initial condition (3).

Remark: The coefficients of t inside the series (17), $n\pi c/l$, are called the frequencies. The smallest frequency, $\pi c/l$, is the fundamental note, while the double, triple, and so on of the fundamental note are the overtones. Notice that by shortening the length l of the vibrating portion of the string with a finger, a violinist produces notes of higher frequency.

$n\lambda/L$ wavenumber

§4.1 Separation of Variables–Example

Example A (not in the textbook)

Use the method of separation of variables to obtain a solution of the wave equation:

$$u_{tt} = u_{xx}, \quad x \in (0, 1),$$

$L = 1$

subject to the conditions:

$c = 1$

$$u(0, t) = u(1, t) = 0,$$
$$u(x, 0) = \sin \pi x + \sin(2\pi x), \quad u_t(x, 0) = 0.$$

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Solution: We may repeat the previous solution process. Here, we just directly use the formula in Theorem 1 with $c = l = 1$, and the solution is

$$u(x, t) = \sum_{n=1}^{\infty} (A_n \cos n\pi t + B_n \sin n\pi t) \sin n\pi x,$$

which satisfies the equation and the homogeneous boundary conditions.

$$u(x, 0) = \sum_{n}^{\infty} A_n \sin n\pi x$$

§4.1 Separation of Variables–Example (Cont'd)

We now use the initial conditions to determine A_n and B_n. Since

$$u(x,0) = \sum_{n=1}^{\infty} A_n \sin n\pi x = \sin \pi x + \sin(2\pi x),$$

we obtain $A_1 = A_2 = 1$ and $A_n = 0$ for all $n \geq 3$.

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By $u_t(x,0)$, we find

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Therefore, the solution is

$$u(x,t) = \cos \pi t \sin \pi x + \cos 2\pi t \sin 2\pi x.$$

This ends the solution process.